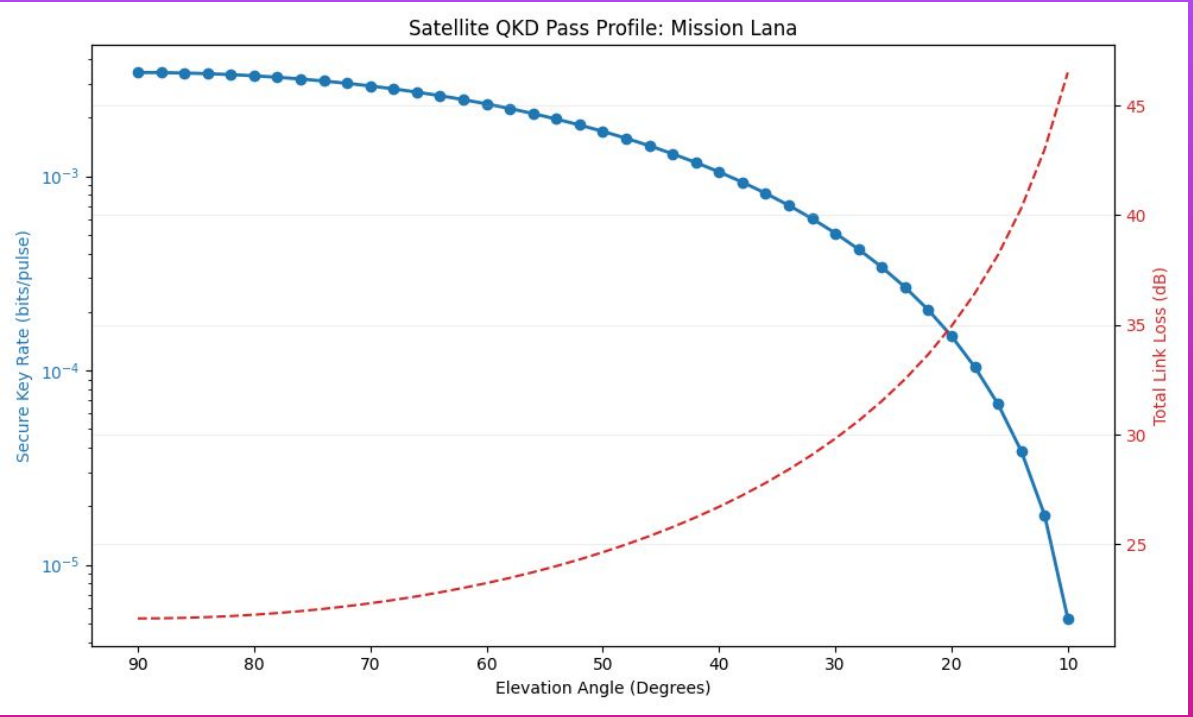
Mission Lana: Satellite Quantum Communication & Teleportation

Simulating Secure Key Distribution in a Noisy LEO Environment

The Pass Profile (Physics & Geometry)

Finding: Link loss scales from 21dB (Zenith) to 46dB (Horizon).

Key Concept: Slant range and atmospheric "air mass" dictate link efficiency.

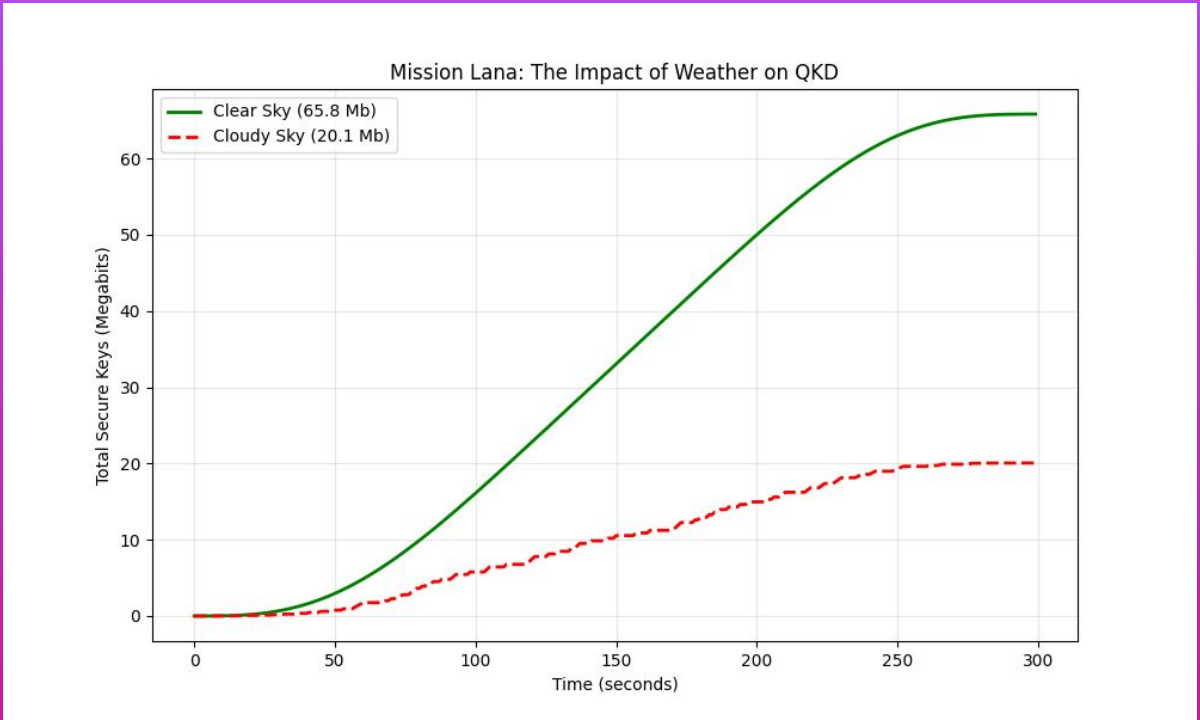


As the elevation angle drops, the atmosphere effectively 'thickens,' killing the quantum signal.

Operational Reality (Weather & Eavesdropping)

Key Concept: QKD is an 'All-or-Nothing' security game.

Finding: Cloud cover reduced our yield by nearly 70% (65Mb down to 20Mb).

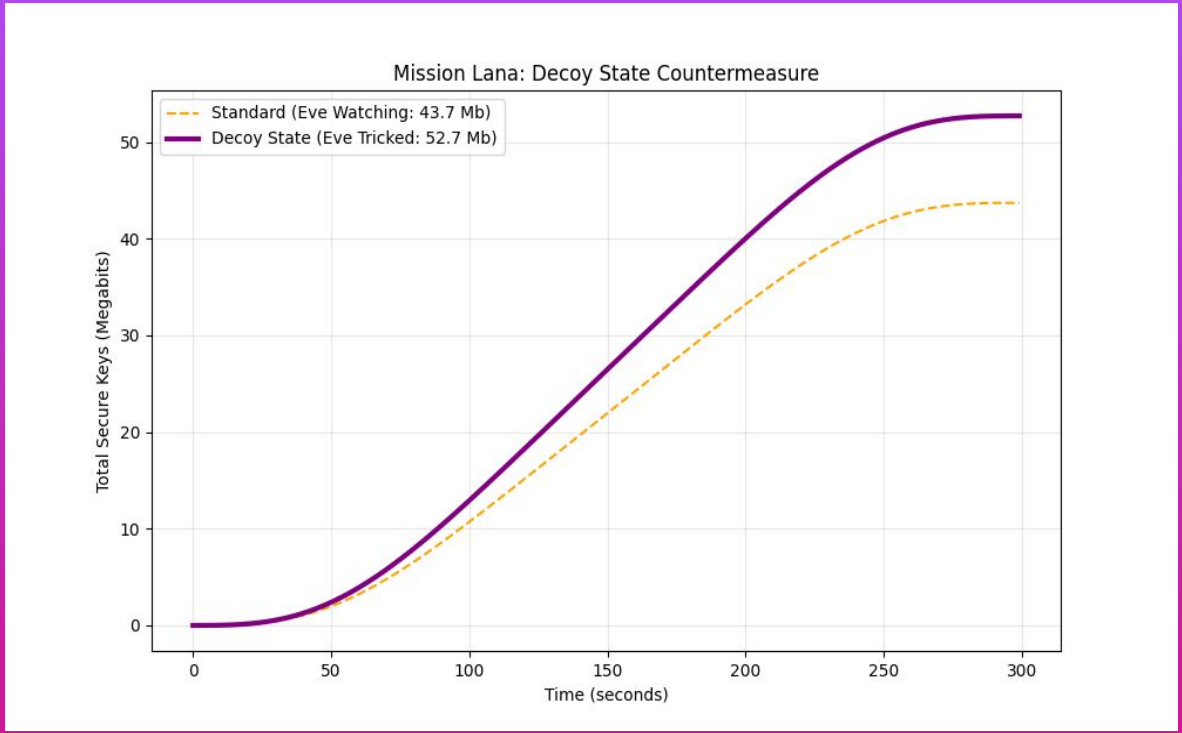


This proves why we need multiple ground stations; a single cloudy night shouldn't stop a defense mission.

The Sneakster & The Countermeasure

Key Concept: Detecting 'Eve' and fighting back with Decoy States.

Finding: Eve's 10% interception dropped us to 43Mb, but Decoy States recovered us back to 52Mb.

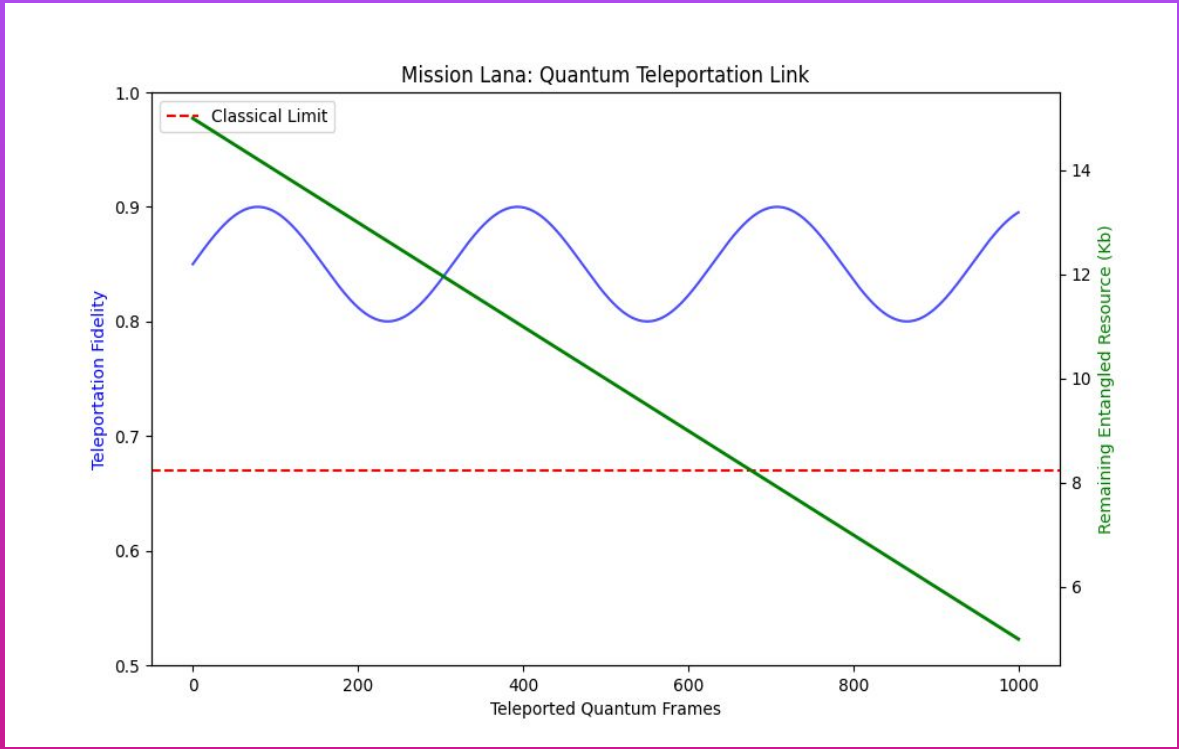


We don't just detect the spy; we use physics to trick her and reclaim our secure bandwidth.

The Quantum Endgame (Entanglement & Teleportation)

Key Concept: Moving beyond lasers to 'Spooky Action.'

Finding: Maintained teleportation fidelity (0.85) well above the 0.67 classical limit.



By sharing entanglement, we teleported 1,000 frames of data with a fidelity that classical computers can't replicate.

Lana Waddell: *Quantum Systems & Aerospace Simulation Developer*

Core Competencies: Python-based physical modeling, Orbital Link Budgeting (SatQuMA), and Quantum Key Distribution (QKD) architecture.

Project Focus: Developing resilient communication links in Low Earth Orbit (LEO) environments using decoy-state protocols and entangled photon distribution.

Technical Environment: High-performance simulation on macOS (M1 architecture) utilizing virtualized Python environments and `matplotlib` for mission-critical data visualization.